
BALAJI RAMACHANDRAN

IoT SOLUTIONS ENGINEER

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PROFESSIONAL SUMMARY

A Research Engineer with nearly 4 years at DEWA's R&D Centre and 8+ years of total experience in IoT, Edge AI, and Software Engineering. Published 10 peer-reviewed papers across IEEE and ASME international conferences — including a Best Paper Award at EECIS 2024 — with 4 papers in progress, reflecting a continuously active research pipeline. Led the end-to-end R&D and development of the OmniHub IoT Terminal, recognised with the Nujoom Award 2024 (DEWA's internal Best Technical Project award) and Engineering Product of the Year (external international recognition), alongside 15+ IoT use cases spanning energy, smart infrastructure, and sustainability. Recipient of the Best Collaborator Award and Quality Achievement Award (2025), with experience mentoring 10+ engineers and representing DEWA at international events. Now transitioning to Senior Researcher, bringing proven capability in independent research leadership, cross-functional collaboration, and delivering R&D outcomes of measurable organisational impact.

SKILLS

Programming: Python, Java, PHP, .NET, C++

Web and Mobile : HTML5, CSS, JavaScript, Android

Data Analysis: Pandas, NumPy, OpenCV, Scikit-learn, Matplotlib, Seaborn, Streamlit, Plotly, D3.js

Frameworks and Libraries: Flask, FastAPI, TensorFlow, Keras, PyTorch, TFLite, Selenium, Postman.

Hardware Platforms: Raspberry Pi, Arduino, IIOT gateways, Linux, Jetson Nano

Cloud and DevOps: AWS, Azure, GCP, Docker, Kubernetes, GitHub, Grafana, Jenkins

Machine Learning: AI frameworks, edge-based AI solutions, LLM's chatbots (OpenAI).

R&D: Research methodology, Technical writing, Prototype development and testing

IIoT & Edge Systems: Industrial IoT platforms (ThingsBoard, Azure IoT, Cumulocity), MQTT/Modbus integration, edge gateways, secure device–cloud architecture.

Vertical Expertise: Energy & Utilities | Industrial IoT & Smart Infrastructure | Telecom & Digital Services | Government & Public Sector | Fintech & Emerging Technologies

EDUCATION

Middlesex University - United Arab Emirates

Master of Science: Engineering Management

Sep 2018 - Dec 2019

Result: First Class with Distinction

Anna University - India

B.E: Electrical and Electronics Engineering,

Aug 2012 - Apr 2016

Result: First class with Distinction

EXPERIENCE

Research Engineer,

Dubai Electricity and Water Authority(DEWA)

Jul 2022- Present

Dubai, UAE

- Conducted applied R&D across 15+ edge AI and IoT use cases for the utility sector — spanning feasibility studies, prototype development, field validation, and production deployment — contributing directly to DEWA's digital innovation agenda.
- Led the full R&D lifecycle of the OmniHub IoT Terminal, from concept and research through to a production-deployed, award-winning multi-protocol device.
- Published 10 peer-reviewed papers in IEEE and ASME international conferences on edge AI, IoT architectures, satellite communication, power quality monitoring, and energy forecasting.
- Established IIoT standards, cybersecurity controls, and data integration frameworks, ensuring research outcomes translate into secure, scalable, and industry-compliant deployments.

- Served as technical research lead bridging business stakeholders, OEM vendors, and engineering teams translating operational KPIs into system-level research proposals, risk assessments, and vendor evaluation frameworks.
- Mentored 10+ engineers and interns in IoT and edge AI research methodologies, and represented DEWA at international events.
- 3 internal invention disclosures submitted, reflecting active contribution to the organisation's intellectual property pipeline.

Software Developer
Future Trend LLC

Jan 2019- Dec 2021
Dubai, UAE

- Executed projects related to Employee Attendance deployed and used by multiple government ministries of UAE.
- Worked as a Scrum Master to deliver Enterprise Projects as part of an Agile Team.
- Developed Backend applications to integrate SDKs of modern facial recognition and biometric devices.
- Provide Technical Leadership to the development Team.

Senior System Engineer
Infosys Limited

Jun 2016 - Aug 2018
Trivandrum, India

- Implemented development, validation, and support activities in line with architecture requirements for various software tools related to automation and web development.
- Designed, developed, and deployed backend services with high availability and scalability.
- Developed web applications that handle high volumes of data and backend services.
- Applied both test-driven and behavior-driven development by following the Software Testing Lifecycle.
- Emphasized cloud deployment and DevOps practices to enhance the efficiency and reliability of software delivery processes.

PROJECTS

Currently Ongoing (2025 – Present) (DEWA)

- **Satellite Image based Survey through AI:** AI-driven satellite image analysis pipeline to classify construction activity across Dubai, with parcel-level GIS integration and end-to-end model deployment framework.
- **SDI Water Quality Monitoring Station :** Integrated dashboard for three sensing stations (SCAN, Algae Online, Coliminder) using OmniHub IoT Terminal with LoRa and LTE connectivity.
- **EFI Fault Monitoring :** Developed 5 field-ready prototypes for real-time Earth Fault Indicator monitoring in substations using OmniHub IoT Terminal, currently under field evaluation.
- **DEWA Borewell Monitoring System :** Developing a customised water level sensor for deployment across 400 borewells in Dubai, eliminating manual inspection entirely.
- **Low-Power IoT Water Metering Terminal :** Designed and deployed an end-to-end M-Bus water meter reading solution transmitting data via NB-IoT, enabling real-time monitoring.

2025 (DEWA)

- **Smart IoT Device with Mobile App :** Field crew smart device integrating thermal cameras, Bluetooth measurement tools, and AI anomaly detection, reducing substation condition monitoring time from 30 to 15 minutes. Recognised by the business unit for operational impact and currently transitioned for further product enhancement and scaling
- **Smart Crew Dispatch :** Built a centralized simulation dashboard providing near real-time visibility of field crew status and job tracking across operations. Implemented automated notifications and email alerts to improve task allocation and dispatch efficiency.
- **OmniHub Certification & Commercialisation :** Led end-to-end CE certification with the vendor, ensuring compliance with international electromagnetic and safety standards. Currently driving commercial market entry in partnership with Infracore, scaling OmniHub beyond DEWA's internal deployment.

- **SGS Power Harmonic Filter Monitoring Dashboard** — Designed an IoT-based dashboard for continuous monitoring of harmonic filters and power quality parameters in substations. Enabled periodic LTE data transmission with automated alerts for abnormal harmonic levels and filter faults.

2024 (DEWA)

- **Energy Disaggregation** — Developed a custom edge device with AI algorithms for high-frequency appliance energy disaggregation at 80% accuracy, deployed in 15 homes with supporting web and mobile applications.
- **Load Forecasting Dashboard** — Python-based web dashboard integrating AI models to forecast feeder loads using 5 years of historical data from 20 substations, with interactive ML metrics and mapping.
- **Decision Support System** — Software architecture for load forecasting and energy KPI analysis using 10,000 customer meter records, enabling performance comparison at region, community, and building levels.
- **Green Hydrogen Monitoring** — Remote dashboard for battery systems at MENA's first Green Hydrogen plant, monitoring Tesla lithium-ion (1.21 MW) and NaS (1.2 MW) state of charge for hydrogen production control.
- **Tesla Powerwall Monitoring and Telemetry System** — Implemented a real-time telemetry system to monitor battery SoC, voltage, current, and temperature of Tesla Powerwall units.
- **Substation Monitoring** — Architecting a scalable solution blueprint to expand real-time telemetry coverage from 15 to 1,000+ substations, covering edge telemetry, data pipeline design, and cloud observability.

2023 (DEWA)

- **OmniHub IoT Terminal** — Led end-to-end development of DEWA's proprietary multi-protocol IoT terminal supporting LoRa Satellite, LoRaWAN, LTE CAT-M, NB-IoT, Wi-Fi, Bluetooth, RS485, and GPIO — solar-capable with custom industrial firmware.
- **3U Nano-Satellite Integration** — Implemented LoRa-based IoT terminal connectivity with DEWA's first nano-satellite DEWASAT-1, demonstrating three utility use cases with ongoing scale-up to additional satellite providers.
- **Power Quality Monitoring** — IoT-based real-time power quality analysis via LTE CAT-M, deployed in substations with automated alarm generation for abnormal events.
- **Smart Lab Infrastructure Monitoring** — IoT-enabled centralised dashboard integrating energy meters, water meters, people counting sensors, and a weather station across DEWA's R&D IoT Lab and 3D Printing Lab, enabling real-time resource tracking and energy saving initiatives.

2019 – 2021 Future Trend LLC, Dubai

- **ERP for SMEs** — Multi-platform ERP application utilised by 10+ SMEs, featuring geo-fencing and payment integration, resulting in a 30% increase in operational efficiency.
- **Attendance App for NOC UAE** — Geo-fenced attendance application with multi-language support, tracking 500+ employees across 3 locations for a UAE government entity.
- **Face Recognition Integration** — Python-based biometric attendance and security system for a semi-government media agency, enabling fast individual recognition for 200+ employees.

2016 – 2018 Infosys Limited, India

- **IPTV Automation** — Developed 100+ automation test scripts for a telecom operator's digital TV services using a TDD model, fully automating the set-top box testing process.
- **Revenue Analytics for Game Developer** — Web application analysing millions in revenue across multiple ad networks, delivering per-game insights and optimising revenue strategies.

PUBLICATIONS

2025

- Ramachandran, B. et al., *Real-Time Power Quality Monitoring in Dubai: A Comparative Study of LoRa Satellite, LoRaWAN, and LTE CAT-M Technologies*, IEEE AICT 2025.
- Dua, G. S., Ramachandran, B. et al., *Smart Crew Dispatch System in Distribution Networks: A Digital Simulation Approach*, IEEE ICITCOM 2025.
- Vohra, M., Ramachandran, B. et al., *Web-Controlled GPR System for Remote Data Acquisition and Surface Reflection Suppression*, IEEE AICT 2025.

2024

- Ramachandran, B. et al., *IoT Enabled Power Quality Monitoring in Substations through LoRa LEO Satellite Communication*, IEEE EECCIS 2024. 🏆 **Best Paper Award**
- Ramachandran, B. et al., *Enhancing Distribution Network Reliability through IoT-Driven Alarm Systems*, IEEE ICSPIS 2024.
- Chikte, R., Ramachandran, B. et al., *Short-Term Time Series Energy Forecasting for Smart Buildings Using Machine Learning Models*, ASME IMECE 2024.

2023

- Ramachandran, B. et al., *Improvement of Direct Communication to Satellite Using LoRa-FHSS Compared to LoRa-CSS (DEWASAT-1)*, IEEE MTTW 2023.
- Ramachandran, B. et al., *IoT-Based Test Facility for a Water Transmission Line with Leak Simulation*, IEEE ITIKD 2023.
- Subramanian, V., Ramachandran, B. et al., *Hardware Doppler Shift Emulation and Compensation for LoRa LEO Satellite Communication*, IEEE ITIKD 2023.
- Haitaamar, Z. N., Ramachandran, B. et al., *Design, Development, and Implementation of IoT-Enabled Laboratory*, IEEE ITIKD 2023.

CERTIFICATIONS

Scrum Master

Oct 2021

AWARDS

Engineering Product of the Year(2026): Digital Engineering Award - Omnihub IoT Terminal
Best Collaborator Award (2025): R&D Center – Cross-department Collaboration
Quality Achievement Award (2025): CE Certification – IoT Terminal
Nujoom Award (2024) : Best Technical Project – Omnihub IoT Terminal
Best Paper Award(2024): PQA Monitoring – EECCIS Conference
Best Ideation Session Award (2024): Innovation & Invention Disclosures

PERSONAL DETAILS

Date of Birth: 15/06/1995

Nationality: Indian

Visa Status: UAE Golden Visa

Languages: English and Tamil